Class 24

Health, Environment, and Externalities

Part IV — Governing Capitalism's Consequences

The American Economy — draft course materials (proposal under discussion)

The point of today's class

Many of capitalism's biggest costs and benefits never pass through market prices — pollution, disease, safety, longevity. Economics gives us tools to measure those externalities and design institutions that reduce harms without pretending regulation is free.

Why it matters

- A factory produces \$1 million of goods and emits pollution that causes asthma, missed school, and premature deaths.
  - GDP records the production.
  - When people then buy inhalers and hospital visits, GDP rises *again*.
- GDP counts the ambulance ride, but not the accident avoided.
- Some of the most valuable things a society buys — clean air, disease prevention, extra years of life — are never sold in a market at all.
- Today, the course's first lesson (GDP is not welfare) gets its final policy application.

Today

1. Externalities: when prices miss the point
2. The policy toolkit — and when bargaining works
3. Clean air: an American success story (and climate, the hard case)
4. Why health care is not a normal market
5. The U.S. health puzzle: more spending, shorter lives
6. Valuing life: should economists price mortality risk?

Externalities: when prices miss the point

- An externality: one person's choice imposes costs or benefits on others that market prices do not capture.

Type	Example	Market problem
Negative	Air pollution	Too much: private cost $<$ social cost
Positive	Vaccination	Too little: private benefit $<$ social benefit
Congestion	Traffic	Each driver slows all the others
Climate	CO_2 emissions	Global, long-lived, diffuse harms

- The one diagram to know: private marginal cost sits below social marginal cost, so the market produces too much pollution-intensive output — unless policy makes the polluter face the full cost.

The policy toolkit — "regulation" is not one thing

Tool	How it works	Best when
Pigouvian tax	Price the harm	Damage per unit is estimable
Cap-and-trade	Fix quantity, trade permits	The quantity target matters
Command-and-control	Mandate a standard	Monitoring emissions is hard
Liability	Polluter pays damages	Harm is traceable to a source
Information	Labels and disclosure	Buyers and firms can respond
Subsidy	Reward positive spillovers	Vaccines, R&D, clean tech

- Coase (Class 21 callback): with clear property rights and low transaction costs, parties can bargain — the rancher and the farmer settle the fence themselves.
- But millions of people breathing polluted air cannot bargain with every smokestack. High transaction costs are where taxes, permits, and standards come in.

Air pollution falls while GDP rises

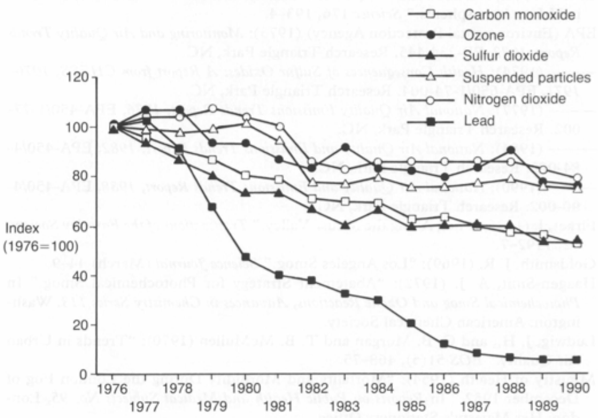


Figure 45.4 Ambient air pollutant concentrations, USA, 1976–90
Source: US Environmental Protection Agency, National Air Quality and Emissions Trends Report, annual.

Source: U.S. Environmental Protection Agency, Our Nation's Air 2025. Six common pollutants down 79% since 1970.

Clean air: how America changed the rules of production

- Since 1970, the six common air pollutants are down 79% — while the economy grew severalfold. The U.S. did not choose between growth and clean air; it changed the rules of production.
- What did it: tailpipe standards and catalytic converters, the lead phaseout, power-plant scrubbers, EPA monitoring and enforcement, cheaper abatement technology, and voters who demanded it.
- EPA's central estimate for the 1990 Clean Air Act amendments: benefits exceed costs by more than 30 to 1 — mostly avoided deaths and illness that never appear in GDP.
- Caution: this is not automatic. Local pollutants fall when rich societies regulate; **climate** is harder — global, long-lived, intergenerational, and beyond any one country's rules.

Why health care is not a normal market (Arrow)

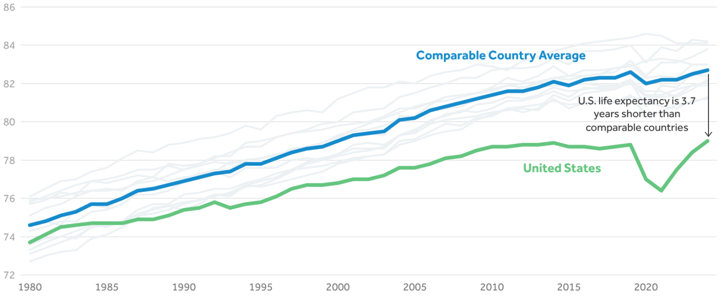
Normal market assumption	Health-care reality
Consumers know what they want	Patients often do not know the diagnosis
Consumers compare prices	Emergencies and insurance obscure prices
Buyers and sellers share information	Doctors know more than patients
Consumers bear the full cost	Insurance pays much of the bill
Bad purchases are reversible	Mistakes can be fatal

- Uncertainty, asymmetric information, insurance, moral hazard, adverse selection, life-or-death stakes: ordinary consumer choice breaks down.
- Health is both a consumption good and human capital — and trust is part of the production function.

The U.S. spends more on health, but lives shorter lives

In 2024, Life Expectancy in the U.S. Reached an All-time High of 79 but Remained Years Behind the Average in Comparable Countries

Life expectancy at birth, in years, 1980-2024



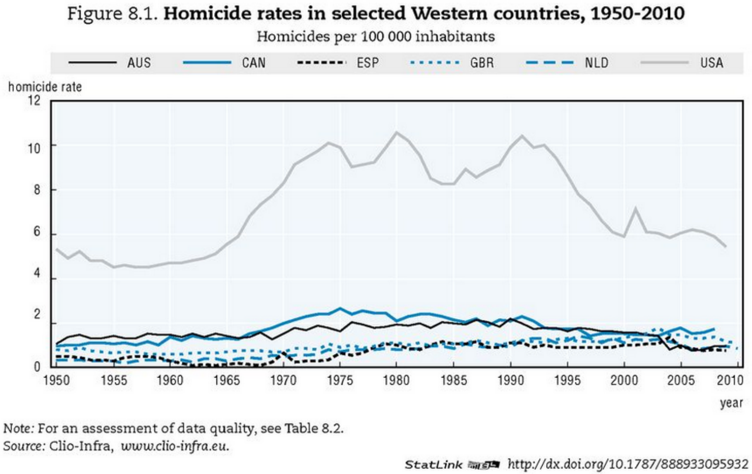
Notes: Comparable countries include Australia, Austria, Belgium, Canada, France, Germany, Japan, the Netherlands, Sweden, Switzerland, and the U.K. See Methods section of "How does U.S. life expectancy compare to other countries?"

Source: KFF analysis of life expectancy data from the OECD and official health and statistics department websites.

Peterson-KFF Health System Tracker

Source: Peterson Center on Healthcare and KFF, Health System Tracker.

The U.S. is not converging to other rich countries



Source: Author's slide (reused from earlier 1740 deck), based on OECD and OWID data.

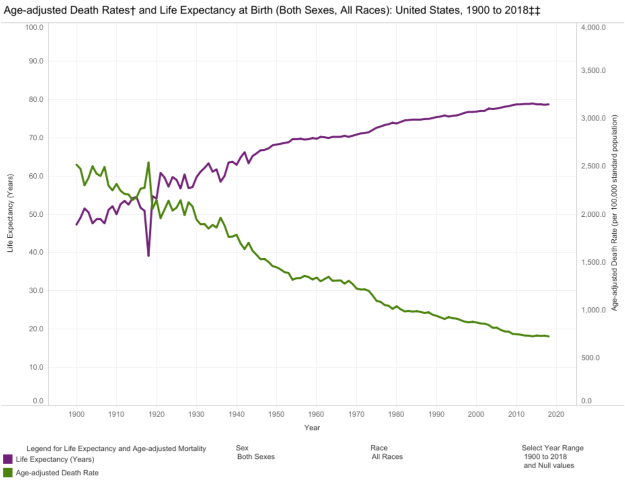
The U.S. health puzzle: why?

- In 2024, U.S. life expectancy reached 79.0 years — about 3.7 years below comparable countries — despite spending far more.
- Because medical care is only one input into health:

Candidate explanation	Mechanism
High prices, admin complexity	More spent per service, not more health
Uneven access	Uninsurance, underinsurance, geography
Chronic disease	Obesity, diabetes, cardiovascular disease
Violence, crashes, overdoses	Deaths of the young weigh heaviest
Inequality and mistrust	Health gradients by income, race, place

- Callback to Class 16: institutional betrayal (Tuskegee) created lasting medical mistrust — trust is health capital.

The rise — and recent U.S. stall — in life expectancy



Source: Our World in Data; CDC National Center for Health Statistics.

Valuing life: the value of a statistical life

- Policy already trades money against mortality risk — every speed limit, drug approval, and pollution standard implies a price on risk. The question is whether to make it explicit.
- The VSL is *not* the value of a named person's life. It sums many people's willingness to pay for small reductions in risk:
  - If 100,000 people each pay $130 to cut a 1-in-100,000 death risk, one statistical life is saved — valued at $13 million.
- HHS's 2024 central estimate: **$13.1 million** per statistical life. QALYs add the quality and length of life-years.
- Cost-benefit analysis is not a way to avoid moral judgment — it is a way to make tradeoffs explicit.

Discussion

Should economists put a dollar value on life? A proposed regulation costs $4 billion and is expected to prevent 200 deaths. Approve it? Now suppose it costs $40 billion. If you refuse to name a value, how do you decide — and who ends up deciding instead?

Read before next class

- Arrow (1963), "Uncertainty and the Welfare Economics of Medical Care" (skim the opening sections).
- EPA, *Our Nation's Air* — browse the charts on emissions and growth.